

D-ary Codes

Problem ID: problem-d-ary-codes

You have found a repository of classic texts from the planet Earth, which you wish to share with others. Unfortunately for you, digital storage devices are rare and expensive, so you have very limited storage space. Additionally, while the source texts are all encoded in Unicode (UTF-8) using binary 0's and 1's, you have a disparate collection of storage devices which use multiple-valued logic. For example, a ternary computer drive (base-3) uses ternary digits or 'trits', where each value can be 0, 1 or 2.

Can you find the optimal variable-length code for compressing a given text using d distinct symbols?

Input

The first line contains two integers n and d , where n is the number of lines in the text, and d is the number of symbols used by your storage drive. Each of the next n lines are strings of Unicode characters, where each line contains a maximum of 80 characters, inclusive of the newline character '\n'.

Problem bounds are $1 \leq n \leq 10000$, and $3 \leq d \leq 100$.

Note: On the inferior Windows operating system, line endings are represented by two characters, '\r\n'.

To avoid any operating system-specific issues, you should ignore any carriage return characters '\r'.

Output

Output the length of the encoded text using an optimal variable-length code for an alphabet of size d .

Sample Input 1

```
1 3
Fair is foul, and foul is fair.
```

Sample Output 1

```
73
```

Sample Input 2

```
2 3
Remember, that I am thy creature: I ought to be thy Adam; but I am rather the
fallen angel, whom thou drivest from joy for no misdeed.
```

Sample Output 2

```
374
```